Lesson 27: Gradient Descent and Convergence Rates

AI/ML Mathematics

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1 Introduction

Gradient descent is the foundational algorithm for continuous optimization. Given a differentiable objective $f : \mathbb{R}^n \to \mathbb{R}$, the method iteratively moves in the direction of steepest descent: $x_{k+1} = x_k - \eta \nabla f(x_k)$. Despite its simplicity, gradient descent—and its stochastic variants—underlies virtually every training algorithm in modern machine learning, from linear regression to deep reinforcement learning.

This lesson establishes precise convergence guarantees for gradient descent under two structural assumptions: L-smoothness (Lipschitz continuity of the gradient) and μ-strong convexity. Section 2 defines L-smoothness and derives its key characterizations. Section 3 proves the *descent lemma*, which provides a quadratic upper bound on the function along any step. Section 4 states the gradient descent algorithm. Section 5 proves the $O(1/k)$ convergence rate for convex L-smooth functions. Section 6 introduces μ-strong convexity. Section 7 proves linear convergence with rate $1 - \mu/L$ for strongly convex functions. Section 8 presents worked examples, and Section 9 provides exercises.

2 L-Smoothness

Definition 2.1 (L-smoothness)**.** A differentiable function $f : \mathbb{R}^n \to \mathbb{R}$ is *L-smooth* (or has *L-Lipschitz continuous gradient*) for some $L > 0$ if

$$\|\nabla f(x) - \nabla f(y)\| \le L \|x - y\| \quad \forall x, y \in \mathbb{R}^n. \tag{1}$$

Remark 2.2. We use $\|\cdot\|$ for the Euclidean norm throughout this lesson unless otherwise stated. The constant L is sometimes called the *smoothness parameter* or *Lipschitz constant of the gradient.*

Proposition 2.3 (Equivalent characterizations of L-smoothness)**.** *For a convex, differentiable function $f : \mathbb{R}^n \to \mathbb{R}$, the following are equivalent:*

(i) f is L-smooth.

(ii) For all $x, y \in \mathbb{R}^n$: $f(y) \le f(x) + \nabla f(x)^\top (y - x) + \frac{L}{2}\|y - x\|^2$.

(iii) For all $x, y \in \mathbb{R}^n$: $(\nabla f(x) - \nabla f(y))^\top (x - y) \ge \frac{1}{L}\|\nabla f(x) - \nabla f(y)\|^2$.

Proof. (i) $\Rightarrow$ (ii): This is the descent lemma, proved in Theorem 3.1 below.

(ii) $\Rightarrow$ (iii): Apply (ii) twice—once with the roles of x and y as stated, and once with x and y swapped:

$$f(y) \le f(x) + \nabla f(x)^\top (y - x) + \frac{L}{2}\|y - x\|^2,$$
$$f(x) \le f(y) + \nabla f(y)^\top (x - y) + \frac{L}{2}\|x - y\|^2.$$

Adding these two inequalities:

$$0 \le (\nabla f(x) - \nabla f(y))^\top (y - x) + L\|x - y\|^2,$$

which rearranges to $(\nabla f(x) - \nabla f(y))^\top (x - y) \ge -L\|x - y\|^2 + 2(\nabla f(x) - \nabla f(y))^\top (x - y)$... Let us proceed more carefully. Adding gives:

$$0 \le [\nabla f(x)^\top (y - x) + \nabla f(y)^\top (x - y)] + L\|x - y\|^2 = -(\nabla f(x) - \nabla f(y))^\top (x - y) + L\|x - y\|^2.$$

Hence $(\nabla f(x) - \nabla f(y))^\top (x - y) \le L\|x - y\|^2$. This gives the *upper* co-coercivity bound.

For the *lower* bound (which is (iii)), define $\varphi(z) = f(z) - \frac{1}{2L}\|\nabla f(z)\|^2$. One can show that φ is convex when f is convex and L-smooth (this is the Baillon–Haddad theorem). From convexity of φ, the co-coercivity inequality follows. We give a direct proof.

Consider the function $h(z) = f(z) - \nabla f(x)^\top z$ for fixed x. This is convex and L-smooth with minimizer at some z^* satisfying $\nabla h(z^*) = \nabla f(z^*) - \nabla f(x) = 0$, i.e., $z^* = x$ (assuming f is strictly convex; for the general case the argument applies to any minimizer). By the quadratic upper bound applied to h:

$$h(y) \le h(x) + \nabla h(x)^\top (y - x) + \frac{L}{2}\|y - x\|^2.$$

Since $\nabla h(x) = 0$: $h(y) \le h(x) + \frac{L}{2}\|y - x\|^2$, giving $f(y) - \nabla f(x)^\top y \le f(x) - \nabla f(x)^\top x + \frac{L}{2}\|y - x\|^2$, which is just (ii) again.

We instead use the following cleaner argument: define $g = f^*$ (the convex conjugate). If f is convex and L-smooth, then g is $(1/L)$-strongly convex. By the gradient inequality for strongly convex functions applied to g, and using the Fenchel–Young identity $\nabla f = (\nabla g)^{-1}$, one obtains (iii). (The full conjugate argument is beyond scope; we accept (iii) as equivalent.)

(iii) $\Rightarrow$ (i): By Cauchy–Schwarz,

$$\frac{1}{L}\|\nabla f(x) - \nabla f(y)\|^2 \le (\nabla f(x) - \nabla f(y))^\top (x - y) \le \|\nabla f(x) - \nabla f(y)\| \cdot \|x - y\|.$$

Dividing both sides by $\|\nabla f(x) - \nabla f(y)\|$ (when nonzero) gives $\|\nabla f(x) - \nabla f(y)\| \le L\|x - y\|$. $\square$

Example 2.4 (Smoothness of quadratics)**.** Let $f(x) = \frac{1}{2}x^\top Ax + b^\top x + c$ with $A \succeq 0$. Then $\nabla f(x) = Ax + b$ and $\|\nabla f(x) - \nabla f(y)\| = \|A(x - y)\| \le \|A\| \cdot \|x - y\|$ where $\|A\| = \lambda_{\max}(A)$. So f is L-smooth with $L = \lambda_{\max}(A)$.

Example 2.5 (Smoothness of logistic loss)**.** The logistic loss $f(w) = \sum_{i=1}^n \log(1 + e^{-y_i x_i^\top w})$ has gradient Lipschitz constant $L = \frac{1}{4}\lambda_{\max}(X^\top X)$, where X is the data matrix with rows $x_i^\top$. This follows from the fact that the second derivative of $\log(1 + e^{-z})$ is bounded above by $1/4$.

3 The Descent Lemma

Theorem 3.1 (Descent lemma – quadratic upper bound)**.** *If $f : \mathbb{R}^n \to \mathbb{R}$ is differentiable and L-smooth, then for all $x, y \in \mathbb{R}^n$:*

$$f(y) \le f(x) + \nabla f(x)^\top (y - x) + \frac{L}{2}\|y - x\|^2. \tag{2}$$

Proof. By the fundamental theorem of calculus applied to the function $\varphi(t) = f(x + t(y - x))$ for $t \in [0, 1]$:

$$f(y) - f(x) = \int_0^1 \nabla f(x + t(y - x))^\top (y - x)\, dt. \tag{3}$$

Subtracting $\nabla f(x)^\top (y - x)$ from both sides:

$$f(y) - f(x) - \nabla f(x)^\top (y - x) = \int_0^1 \big[\nabla f(x + t(y - x)) - \nabla f(x)\big]^\top (y - x)\, dt. \tag{4}$$

Applying the Cauchy–Schwarz inequality to the integrand:

$$\le \int_0^1 \|\nabla f(x + t(y - x)) - \nabla f(x)\| \cdot \|y - x\|\, dt. \tag{5}$$

By L-smoothness of f (Definition 2.1):

$$\|\nabla f(x + t(y - x)) - \nabla f(x)\| \leq L \|t(y - x)\| = Lt \|y - x\|. \tag{6}$$

Substituting (6) into (5):

$$f(y) - f(x) - \nabla f(x)^\top (y - x) \leq \int_0^1 Lt \|y - x\|^2 \, dt = L\|y - x\|^2 \int_0^1 t \, dt = \frac{L}{2}\|y - x\|^2. \tag{7}$$

Rearranging gives (2). □

Descent Lemma

For L-smooth f, the tangent hyperplane plus a quadratic term provides a global upper bound: $f(y) \leq f(x) + \nabla f(x)^\top (y - x) + \frac{L}{2}\|y - x\|^2$. This is the key inequality that drives all convergence analysis for gradient descent.

Corollary 3.2 (Sufficient decrease)**.** *If f is L-smooth and $x^+ = x - \frac{1}{L}\nabla f(x)$, then*

$$f(x^+) \leq f(x) - \frac{1}{2L}\|\nabla f(x)\|^2. \tag{8}$$

Proof. Apply the descent lemma with $y = x^+ = x - \frac{1}{L}\nabla f(x)$:

$$\begin{aligned} f(x^+) &\leq f(x) + \nabla f(x)^\top \Big(-\frac{1}{L}\nabla f(x)\Big) + \frac{L}{2}\Big\|\frac{1}{L}\nabla f(x)\Big\|^2 \\ &= f(x) - \frac{1}{L}\|\nabla f(x)\|^2 + \frac{1}{2L}\|\nabla f(x)\|^2 = f(x) - \frac{1}{2L}\|\nabla f(x)\|^2. \end{aligned}$$

□

Remark 3.3. The step size $\eta = 1/L$ is the largest step size for which the descent lemma guarantees a decrease at every iteration. Larger step sizes may cause the iterates to diverge; smaller step sizes give a smaller per-step improvement. The choice $\eta = 1/L$ is thus optimal for the worst case among L-smooth functions.

4 The Gradient Descent Algorithm

Definition 4.1 (Gradient descent)**.** Given a differentiable function $f : \mathbb{R}^n \to \mathbb{R}$, an initial point $x_0 \in \mathbb{R}^n$, and a step size $\eta > 0$, the *gradient descent* iteration is

$$x_{k+1} = x_k - \eta \nabla f(x_k), \quad k = 0, 1, 2, \ldots \tag{9}$$

Remark 4.2 (Steepest descent interpretation). Each step moves in the direction $-\nabla f(x_k)$, which is the direction of steepest descent of f at x_k in the Euclidean norm. More precisely, $-\nabla f(x_k) = \arg\min_{\|d\|=1} \nabla f(x_k)^\top d$.

Proposition 4.3 (Monotone decrease)**.** *If f is L-smooth and $\eta = 1/L$, then the gradient descent iterates satisfy*

$$f(x_{k+1}) \leq f(x_k) - \frac{1}{2L}\|\nabla f(x_k)\|^2 \leq f(x_k). \tag{10}$$

In particular, $\{f(x_k)\}_{k\geq 0}$ is a non-increasing sequence.

Proof. This is exactly Corollary 3.2 applied with $x = x_k$. □

5 $O(1/k)$ Convergence for Convex L-Smooth Functions

Theorem 5.1 ($O(1/k)$ convergence rate)**.** *Let $f : \mathbb{R}^n \to \mathbb{R}$ be convex and L-smooth with a minimizer x^*. Gradient descent with step size $\eta = 1/L$ satisfies*

$$f(x_k) - f(x^*) \leq \frac{L\,\|x_0 - x^*\|^2}{2k} \tag{11}$$

for all $k \geq 1$.

Proof. We proceed in four steps.

Step 1: Sufficient decrease. By Corollary 3.2 (Proposition 4.3):

$$f(x_{k+1}) \leq f(x_k) - \frac{1}{2L}\|\nabla f(x_k)\|^2. \tag{12}$$

Step 2: Distance contraction. We compute the squared distance to x^*:

$$\begin{aligned}\|x_{k+1} - x^*\|^2 &= \|x_k - \tfrac{1}{L}\nabla f(x_k) - x^*\|^2 \\ &= \|x_k - x^*\|^2 - \frac{2}{L}\nabla f(x_k)^\top (x_k - x^*) + \frac{1}{L^2}\|\nabla f(x_k)\|^2.\end{aligned} \tag{13}$$

By convexity of f:

$$f(x^*) \geq f(x_k) + \nabla f(x_k)^\top (x^* - x_k), \tag{14}$$

which gives $\nabla f(x_k)^\top (x_k - x^*) \geq f(x_k) - f(x^*)$. Substituting into (13):

$$\|x_{k+1} - x^*\|^2 \leq \|x_k - x^*\|^2 - \frac{2}{L}(f(x_k) - f(x^*)) + \frac{1}{L^2}\|\nabla f(x_k)\|^2. \tag{15}$$

Step 3: Eliminate the gradient norm. From Step 1: $\|\nabla f(x_k)\|^2 \leq 2L(f(x_k) - f(x_{k+1}))$. Substituting into (15):

$$\begin{aligned}\|x_{k+1} - x^*\|^2 &\leq \|x_k - x^*\|^2 - \frac{2}{L}(f(x_k) - f(x^*)) + \frac{2}{L}(f(x_k) - f(x_{k+1})) \\ &= \|x_k - x^*\|^2 - \frac{2}{L}(f(x_{k+1}) - f(x^*)).\end{aligned} \tag{16}$$

Rearranging:

$$f(x_{k+1}) - f(x^*) \leq \frac{L}{2}\big(\|x_k - x^*\|^2 - \|x_{k+1} - x^*\|^2\big). \tag{17}$$

Step 4: Telescoping sum. Denote $\delta_k = f(x_k) - f(x^*)$. Summing (17) from $k = 0$ to $k = K - 1$:

$$\sum_{k=1}^{K} \delta_k \leq \frac{L}{2}\big(\|x_0 - x^*\|^2 - \|x_K - x^*\|^2\big) \leq \frac{L}{2}\|x_0 - x^*\|^2. \tag{18}$$

Since δ_k is non-increasing (by Proposition 4.3), $K\,\delta_K \leq \sum_{k=1}^{K} \delta_k$. Hence:

$$\delta_K = f(x_K) - f(x^*) \leq \frac{L\|x_0 - x^*\|^2}{2K}. \tag{19}$$

□

Remark 5.2 (Sublinear rate). The rate $O(1/k)$ is called *sublinear*: to halve the error, we need roughly twice as many iterations. To achieve $f(x_k) - f(x^*) \leq \epsilon$, we need $k \geq L\|x_0 - x^*\|^2/(2\epsilon)$, which is $O(1/\epsilon)$ iterations.

Remark 5.3 (Optimality of the rate). The $O(1/k)$ rate is optimal for gradient descent on the class of convex L-smooth functions. Nesterov showed that no first-order method using only gradient evaluations can achieve a rate better than $O(1/k^2)$ on this class, and his accelerated gradient method achieves this lower bound (covered in Lesson 28).

6 μ-Strong Convexity

Definition 6.1 (μ-strong convexity)**.** A differentiable function $f : \mathbb{R}^n \to \mathbb{R}$ is *μ-strongly convex* for some $\mu > 0$ if for all $x, y \in \mathbb{R}^n$:

$$f(y) \geq f(x) + \nabla f(x)^\top (y - x) + \frac{\mu}{2}\|y - x\|^2. \tag{20}$$

Equivalently, $f(x) - \frac{\mu}{2}\|x\|^2$ is convex.

Proposition 6.2 (Unique minimizer)**.** *If f is μ-strongly convex, then f has a unique minimizer x^*.*

Proof. **Existence:** Since $f(x) \geq f(x_0) + \nabla f(x_0)^\top (x - x_0) + \frac{\mu}{2}\|x - x_0\|^2 \to +\infty$ as $\|x\| \to \infty$, f is coercive. By the extreme value theorem on closed balls and coercivity, a minimizer exists.

Uniqueness: Suppose x^* and y^* are both minimizers with $f(x^*) = f(y^*) = f^*$. By strong convexity at x^*: $f(y^*) \geq f(x^*) + \nabla f(x^*)^\top (y^* - x^*) + \frac{\mu}{2}\|y^* - x^*\|^2$. Since x^* is a minimizer, $\nabla f(x^*) = 0$. Thus $f^* \geq f^* + \frac{\mu}{2}\|y^* - x^*\|^2$, giving $\|y^* - x^*\| = 0$. □

Definition 6.3 (Condition number)**.** If f is μ-strongly convex and L-smooth, the *condition number* is

$$\kappa = \frac{L}{\mu} \geq 1. \tag{21}$$

The condition number governs the convergence speed of first-order methods. A well-conditioned problem (κ close to 1) converges quickly; an ill-conditioned problem ($\kappa \gg 1$) converges slowly.

Lemma 6.4 (Polyak–Lojasiewicz inequality)**.** *If f is μ-strongly convex with minimizer x^*, then for all $x \in \mathbb{R}^n$:*

$$\|\nabla f(x)\|^2 \geq 2\mu\,(f(x) - f(x^*)). \tag{22}$$

Proof. By strong convexity, for all $y \in \mathbb{R}^n$:

$$f(y) \geq f(x) + \nabla f(x)^\top (y - x) + \frac{\mu}{2}\|y - x\|^2.$$

The right-hand side is a quadratic function of y. Minimizing over y: the minimizer is $y^* = x - \frac{1}{\mu}\nabla f(x)$, and the minimum value is

$$f(x) + \nabla f(x)^\top \Big(-\frac{1}{\mu}\nabla f(x)\Big) + \frac{\mu}{2}\Big\|\frac{1}{\mu}\nabla f(x)\Big\|^2 = f(x) - \frac{1}{2\mu}\|\nabla f(x)\|^2.$$

Since $f(x^*) \leq f(y)$ for all y, and in particular for this y^*:

$$f(x^*) \geq f(x) - \frac{1}{2\mu}\|\nabla f(x)\|^2.$$

Wait—we need the inequality in the other direction. Since $f(y^*) \geq f(x) - \frac{1}{2\mu}\|\nabla f(x)\|^2$ and $f(x^*) \leq f(y^*)$ for all y, we get: the quadratic lower bound on $f(y)$ achieves its minimum $f(x) - \frac{1}{2\mu}\|\nabla f(x)\|^2$. Since $f(x^*) = \min_y f(y)$ and $f(y) \geq f(x) - \frac{1}{2\mu}\|\nabla f(x)\|^2$ for the minimizing y^* of the lower bound, we obtain

$$f(x^*) \geq f(x) - \frac{1}{2\mu}\|\nabla f(x)\|^2,$$

which rearranges to $\|\nabla f(x)\|^2 \geq 2\mu(f(x) - f(x^*))$. □

Remark 6.5. The PL inequality is weaker than strong convexity: there exist non-convex functions satisfying (22). When the PL inequality holds, gradient descent still converges linearly, even without convexity. This is relevant for over-parameterized neural networks, where the loss landscape may satisfy a local PL condition near the solution.

7 Linear Convergence for Strongly Convex Functions

Theorem 7.1 (Linear convergence)**.** *Let $f : \mathbb{R}^n \to \mathbb{R}$ be μ-strongly convex and L-smooth with minimizer x^*. Gradient descent with step size $\eta = 1/L$ satisfies*

$$f(x_k) - f(x^*) \leq \left(1 - \frac{1}{\kappa}\right)^k \big(f(x_0) - f(x^*)\big), \tag{23}$$

where $\kappa = L/\mu$ is the condition number. Consequently, $f(x_k) - f(x^*) \leq \epsilon$ after at most

$$k \geq \kappa \ln\left(\frac{f(x_0) - f(x^*)}{\epsilon}\right) \tag{24}$$

iterations.

Proof. **Step 1: Sufficient decrease.** From Corollary 3.2:

$$f(x_{k+1}) \leq f(x_k) - \frac{1}{2L}\|\nabla f(x_k)\|^2. \tag{25}$$

Step 2: Apply the PL inequality. By Lemma 6.4: $\|\nabla f(x_k)\|^2 \geq 2\mu(f(x_k) - f(x^*))$. Substituting into (25):

$$\begin{aligned} f(x_{k+1}) &\leq f(x_k) - \frac{1}{2L} \cdot 2\mu \left(f(x_k) - f(x^*)\right) \\ &= f(x_k) - \frac{\mu}{L}(f(x_k) - f(x^*)). \end{aligned} \tag{26}$$

Step 3: Contraction. Subtracting $f(x^*)$ from both sides of (26):

$$\begin{aligned} f(x_{k+1}) - f(x^*) &\leq f(x_k) - f(x^*) - \frac{\mu}{L}(f(x_k) - f(x^*)) \\ &= \left(1 - \frac{\mu}{L}\right)(f(x_k) - f(x^*)) = \left(1 - \frac{1}{\kappa}\right)(f(x_k) - f(x^*)). \end{aligned} \tag{27}$$

Step 4: Induction. Let $\rho = 1 - 1/\kappa \in [0, 1)$. Applying (27) recursively:

$$f(x_k) - f(x^*) \leq \rho^k \left(f(x_0) - f(x^*)\right). \tag{28}$$

This establishes (23).

Step 5: Iteration count. To achieve $\rho^k(f(x_0) - f(x^*)) \leq \epsilon$, we need $k \geq \frac{\ln((f(x_0)-f(x^*))/\epsilon)}{-\ln \rho}$. Using $-\ln(1 - 1/\kappa) \geq 1/\kappa$ (since $-\ln(1 - u) \geq u$ for $u \in [0, 1)$), we get $k \geq \kappa \ln((f(x_0) - f(x^*))/\epsilon)$. □

Gradient Descent Convergence Summary

For L-smooth functions with step size $\eta = 1/L$:

- **Convex:** $f(x_k) - f^* = O(1/k)$, i.e., $O(1/\epsilon)$ iterations for ϵ-accuracy.
- **μ-strongly convex:** $f(x_k) - f^* \leq (1 - 1/\kappa)^k(f(x_0) - f^*)$, i.e., $O(\kappa \log(1/\epsilon))$ iterations.

The strong convexity rate is *linear* (or *geometric*): each iteration reduces the error by a constant factor $1 - 1/\kappa$.

Remark 7.2 (Terminology). In optimization, "linear convergence" means the error decreases geometrically: $\epsilon_k \leq C\rho^k$ for some $\rho \in (0, 1)$. This is analogous to the convergence of a geometric series. The term "linear" refers to the linear relationship $\log \epsilon_k \leq \log C + k \log \rho$ on a logarithmic scale.

8 Worked Examples

Example 8.1 (Gradient descent on a quadratic)**.** Let $f(x) = \frac{1}{2}x^\top Ax - b^\top x$ where $A = \mathrm{diag}(\lambda_1, \ldots, \lambda_n)$ with $0 < \mu = \lambda_{\min} \le \lambda_{\max} = L$.

The gradient is $\nabla f(x) = Ax - b$ and the unique minimizer is $x^* = A^{-1}b$. The gradient descent update with $\eta = 1/L$ gives:

$$x_{k+1} = x_k - \frac{1}{L}(Ax_k - b) = (I - \frac{1}{L}A)x_k + \frac{1}{L}b.$$

The error $e_k = x_k - x^*$ satisfies $e_{k+1} = (I - \frac{1}{L}A)e_k$, so $e_k = (I - \frac{1}{L}A)^k e_0$. The spectral radius of $I - \frac{1}{L}A$ is $\max_i |1 - \lambda_i/L| = 1 - \mu/L = 1 - 1/\kappa$. Thus $\|e_k\| \le (1 - 1/\kappa)^k \|e_0\|$, confirming the linear rate.

Example 8.2 (Effect of condition number)**.** Consider $f(x_1, x_2) = \frac{1}{2}(x_1^2 + 100x_2^2)$. Here $L = 100$, $\mu = 1$, $\kappa = 100$. Gradient descent converges as $(1-1/100)^k = 0.99^k$: after $k = 100$ iterations, the error is reduced by a factor of about $0.99^{100} \approx 0.37$, i.e., less than one order of magnitude. After $k = 460$ iterations, the error is reduced by a factor of $0.99^{460} \approx 0.01$ (two orders of magnitude).

In contrast, for $f(x_1, x_2) = \frac{1}{2}(x_1^2 + x_2^2)$ with $\kappa = 1$, one step of gradient descent with $\eta = 1$ gives the exact solution: $x_1 = x_0 - \nabla f(x_0) = x_0 - x_0 = 0$.

Example 8.3 (Convex but not strongly convex)**.** Consider $f(x) = |x|$ on $\mathbb{R}$. This is convex but neither smooth (not differentiable at 0) nor strongly convex ($f(x) - \frac{\mu}{2}x^2$ is not convex for any $\mu > 0$ since it becomes concave for large $|x|$). Gradient descent is not directly applicable here; one would use subgradient methods (Lesson 29).

Consider instead the smooth convex function $f(x) = \frac{1}{2}x^2$ on $[0, \infty)$. This is 1-smooth and 1-strongly convex on $\mathbb{R}$. Gradient descent gives $x_{k+1} = x_k - x_k = 0$: one step suffices ($\kappa = 1$).

RL/ML Connection

Policy gradient step sizes. In vanilla policy gradient (REINFORCE), the update $\theta_{k+1} = \theta_k + \eta \widehat{\nabla} J(\theta_k)$ is stochastic gradient *ascent* on the expected return $J(\theta)$. The smoothness constant L of J (or of the surrogate objective) determines the largest stable step size: $\eta \le 1/L$. In practice, L is unknown and step sizes are tuned empirically or adapted via line search.

TD learning and the condition number. Temporal difference learning with linear function approximation performs a form of stochastic semi-gradient descent. The effective condition number depends on the feature matrix and the discount factor γ: as $\gamma \to 1$, the effective condition number $\kappa \to \infty$, slowing convergence. This is a fundamental difficulty in near-undiscounted RL.

Natural gradient. The natural gradient preconditions the gradient by the inverse Fisher information matrix, effectively transforming the parameter space so that the "curvature" is uniform. This reduces the effective condition number from κ toward 1, analogous to using Newton's method instead of gradient descent.

9 Exercises

Exercise 9.1 (Smoothness verification)**.** Show that $f(x) = \frac{1}{4}\|x\|^4$ on $\mathbb{R}^n$ is *not* L-smooth for any finite L. *Hint:* Compute $\nabla f(x)$ and show that $\|\nabla f(x) - \nabla f(y)\|/\|x - y\|$ is unbounded.

Exercise 9.2 (Descent lemma tightness)**.** For the quadratic $f(x) = \frac{L}{2}x^2$, show that the descent lemma bound $f(y) \le f(x) + f'(x)(y - x) + \frac{L}{2}(y - x)^2$ holds with equality for all x, y.

Exercise 9.3 (Strong convexity of ridge regression)**.** Let $f(w) = \frac{1}{2n}\|Xw - y\|^2 + \frac{\lambda}{2}\|w\|^2$ with $X \in \mathbb{R}^{n\times d}$, $y \in \mathbb{R}^n$, $\lambda > 0$. Show that f is μ-strongly convex with $\mu = \lambda$ and L-smooth with $L = \frac{1}{n}\lambda_{\max}(X^\top X) + \lambda$. Compute the condition number κ and determine how many gradient descent iterations are needed for ϵ-accuracy.

Exercise 9.4 (PL inequality without convexity)**.** Let $f(x) = x^2(3 - x)^2$ on $\mathbb{R}$. Show that f satisfies the PL inequality $|f'(x)|^2 \geq 2\mu\, f(x)$ for some $\mu > 0$ in a neighborhood of $x = 0$. Conclude that gradient descent with sufficiently small step size converges linearly to $x = 0$ from nearby initial points.

Exercise 9.5 (Convergence in iterates)**.** Under the assumptions of Theorem 7.1, prove that

$$\|x_k - x^*\|^2 \leq \frac{2}{\mu}\left(1 - \frac{1}{\kappa}\right)^k (f(x_0) - f(x^*)). \tag{29}$$

Hint: Use strong convexity to relate $\|x - x^*\|^2$ to $f(x) - f(x^*)$.

Exercise 9.6 (Optimal step size for quadratics)**.** For $f(x) = \frac{1}{2}x^\top Ax - b^\top x$ with $A \succ 0$, the gradient descent error satisfies $e_{k+1} = (I - \eta A)e_k$. Show that the spectral radius $\rho(\eta) = \max(|1 - \eta\lambda_{\min}|, |1 - \eta\lambda_{\max}|)$ is minimized by $\eta^* = \frac{2}{\lambda_{\min}+\lambda_{\max}}$, giving $\rho^* = \frac{\kappa-1}{\kappa+1}$. Compare this to the rate $\rho = 1 - 1/\kappa$ obtained with $\eta = 1/L$.

Exercise 9.7 (Lower bound argument)**.** Construct a convex L-smooth function $f : \mathbb{R} \to \mathbb{R}$ and an initial point x_0 such that gradient descent with $\eta = 1/L$ satisfies $f(x_k) - f(x^*) \geq \frac{L\|x_0-x^*\|^2}{4(k+1)}$ for all k. This shows the $O(1/k)$ rate is tight up to a constant. *Hint:* Consider $f(x) = \frac{L}{2}x^2$ with $x_0 = 1$.

References